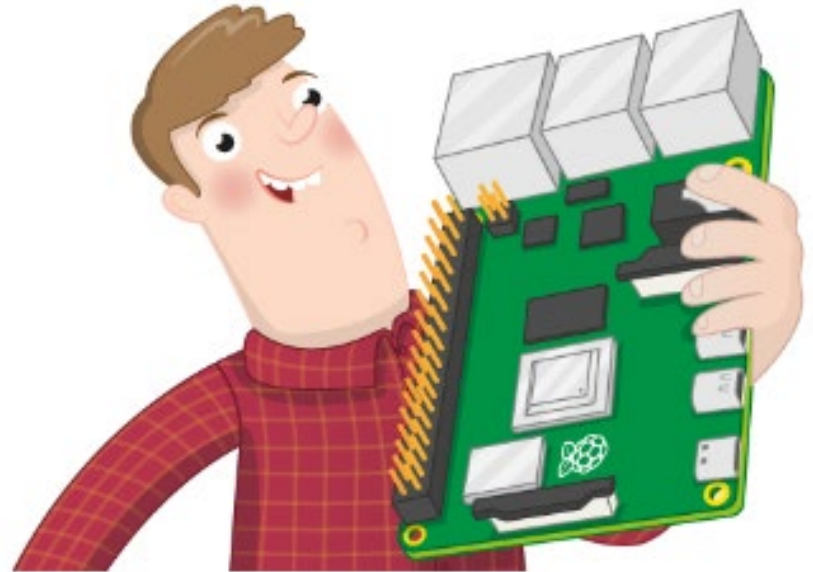


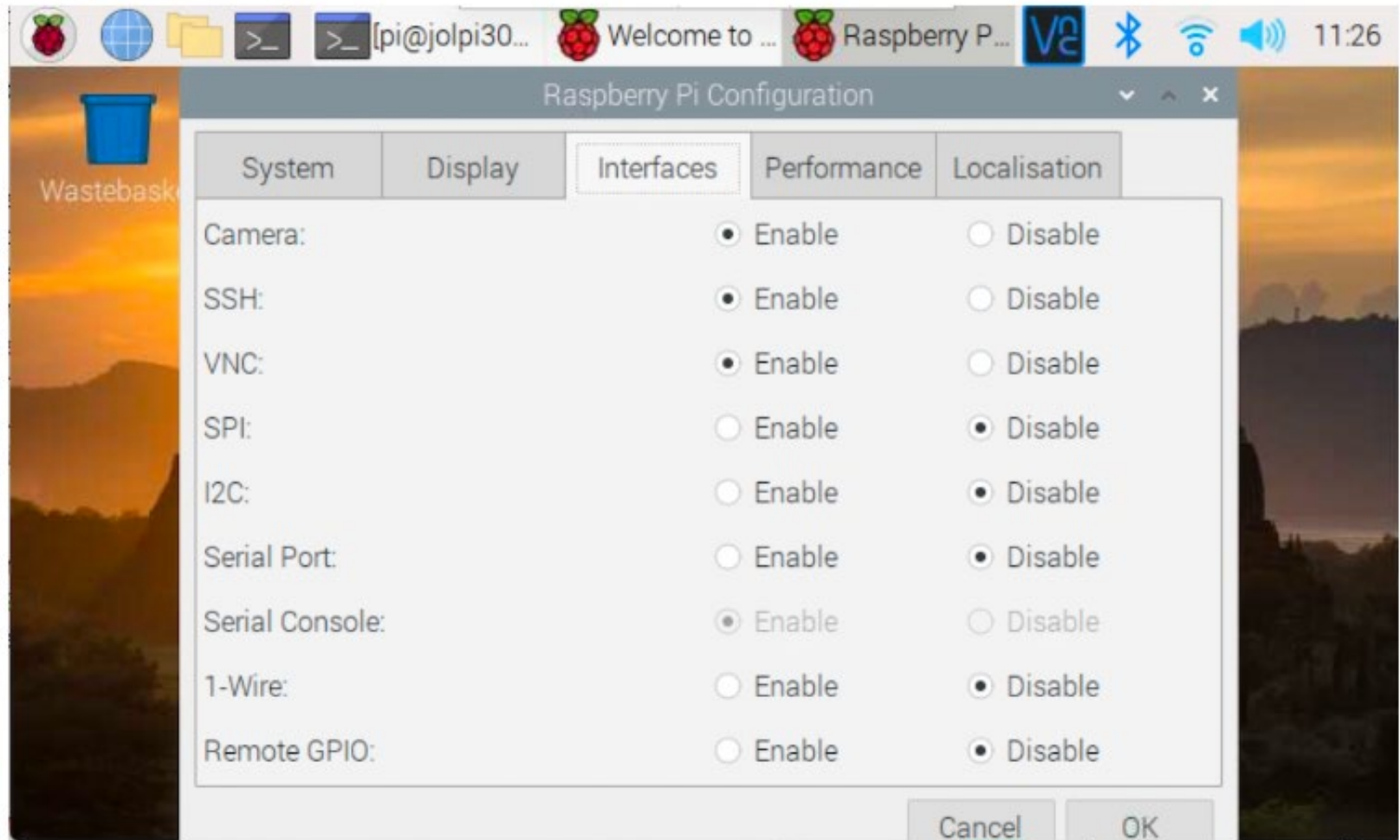
Industrial Internet of Things

Project Raspberry Pi (RPI) & Sensors



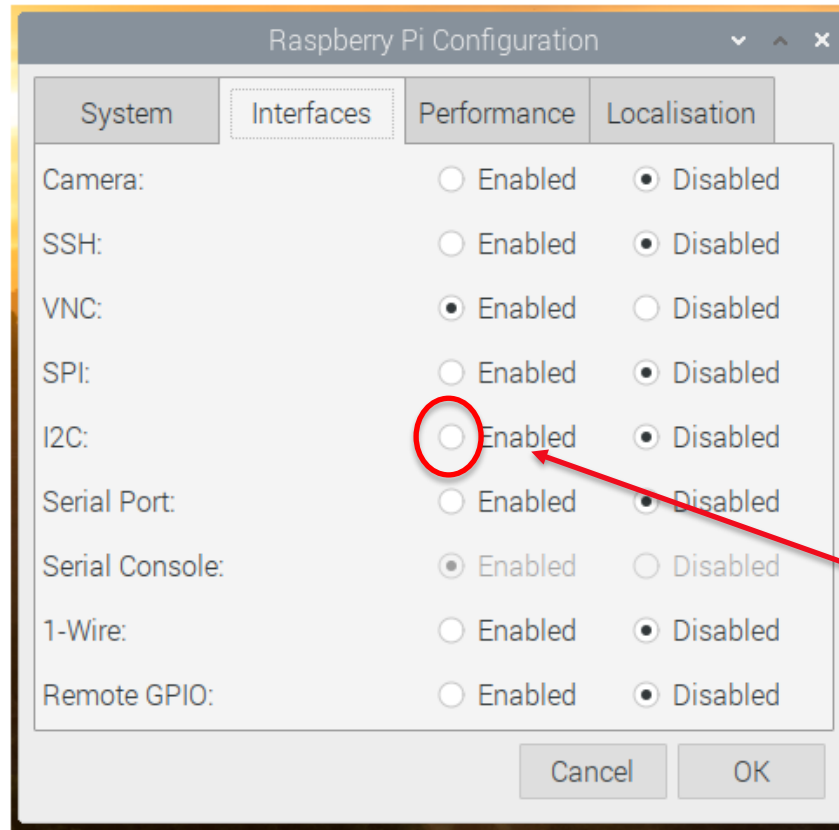
Configuring your RPi

To open the Control Centre, select the Raspberry Pi icon in the top-left corner of the system tray and then go to **Preferences > Control Centre > Interfaces**.



Configuring your Rpi – Enable I2C

You can link devices and components to your Raspberry Pi using a lot of different types of connections. The **Interfaces** tab is where you turn these different connections on or off, so that your Raspberry Pi recognises that you've linked something to it via a particular type of connection.



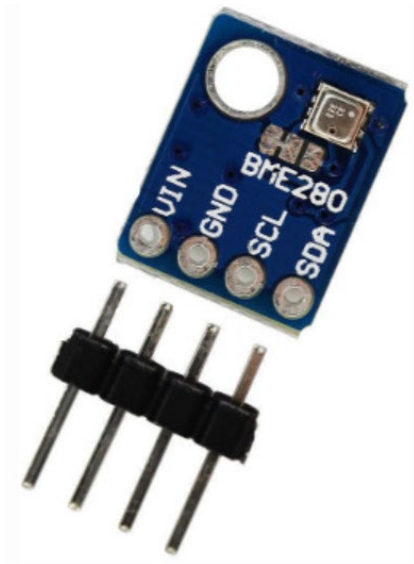
Need to enable I2C
for BME280 sensor

OR `$ sudo raspi-config`

<https://www.raspberrypi.com/documentation/computers/configuration.html>

Temperature/Humidity/Pressure Sensors (BME 280)

Interfacing BME280 digital sensor module (capable of sensing temperature, humidity and pressure) in Python 3 using I2C on the Raspberry Pi.



Board Pin	Name	Remarks	RPi Pin	RPi Function
1	VIN	+3.3V Power	P01-1	3V3
2	GND	Ground	P01-6	GND
3	SCL	Clock	P01-5	GPIO 3 (SCL)
4	SDA	Data	P01-3	GPIO 2 (SDA)

<https://pypi.org/project/RPi.bme280/>

Installing BME 280 in Rpi (Trixie OS)

From Rpi terminal

create your project directory, such as <my_project>

\$ mkdir **my_project** && cd **my_project**

create your virtual environment, such as <env>

\$ python3 -m venv **env**

active virtual environment

\$ source env/bin/activate

install BME 280

(env) \$ pip3 install **RPi.bme280**

to deactivate virtual environment

\$ deactivate

[python-virtual-environment-usage-on-raspberry-pi](#)

Testing the BME 280

- Checking the port and address

```
$ i2cdetect -y 1
          0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:      -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- -- -- -- -- -- -- --
```

Port

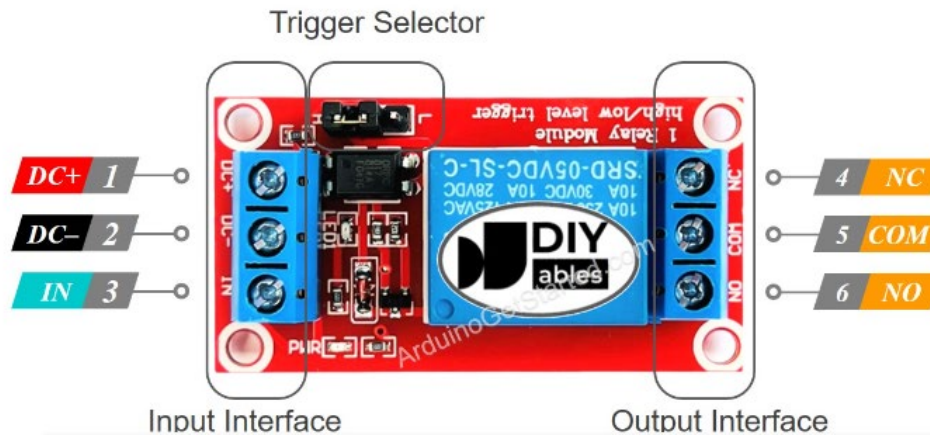
Address

- Run your python script
<https://pypi.org/project/RPi.bme280/>

Relay and 5V DC Fan

A relay is a programmable electrical switch, which can be controlled by Arduino or any micro-controller. It is used to programmatically control on/off the devices, which use the high voltage and/or high current.

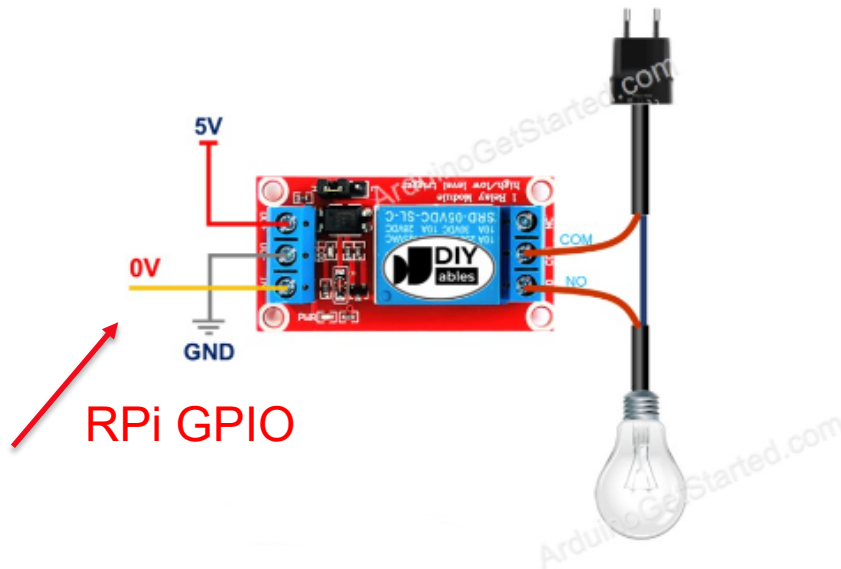
Relay Pinout



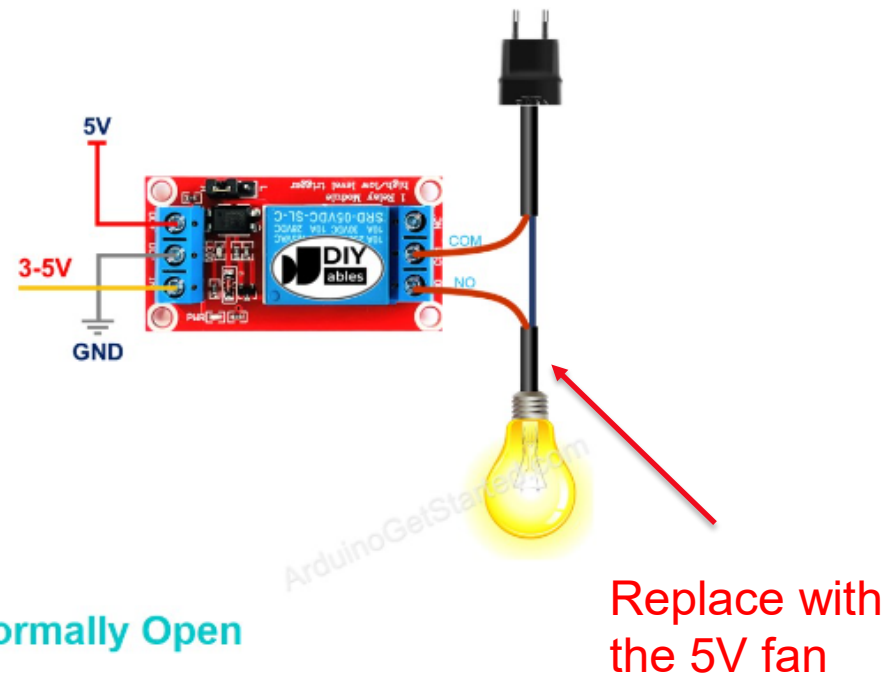
Relay and 5V DC Fan

To use this mode, we need to connect the high voltage device to the COM pin and NO pin.

- ◆ If the IN pin is connected to **LOW** (0V), the switch is open. The device is **OFF** (or inactive).
- ◆ If the IN pin is connected to **HIGH** (5V), the switch is closed. The device is **ON** (or active).



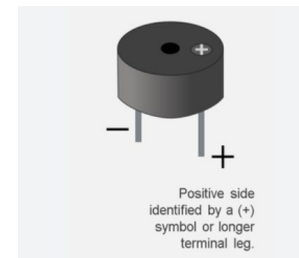
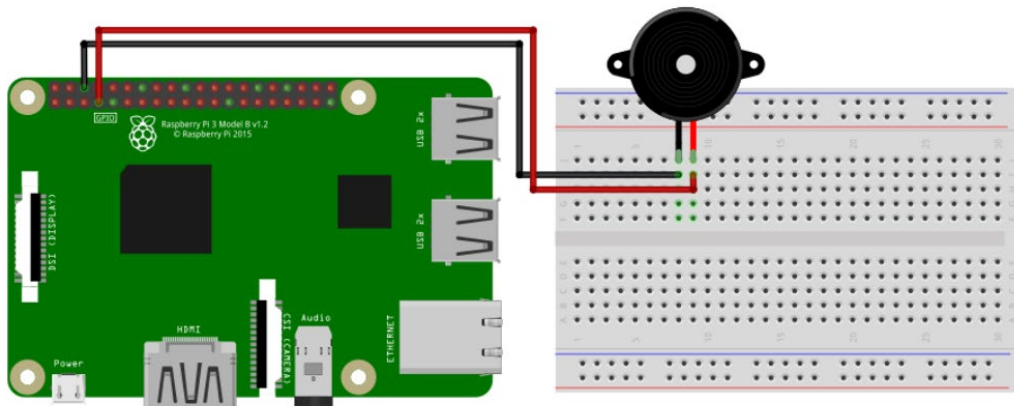
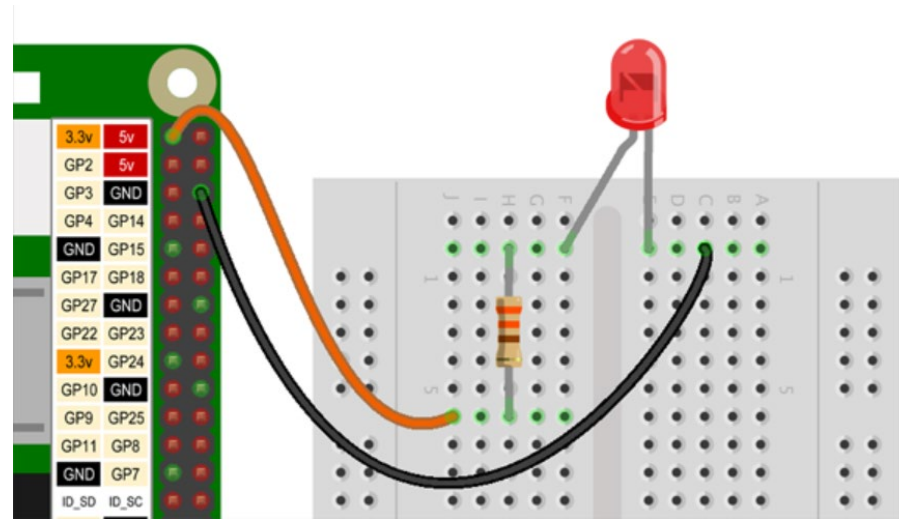
HIGH Level Trigger - Normally Open



<https://arduinogetstarted.com/tutorials/arduino-relay>

LED and Buzzer

Lab 7 Exercise, toggle the LED



Buzzer

Checking your IP address

To check IP address

- `ifconfig`

To install Vim in your RPi

- `sudo apt-get update`
- `sudo apt-get install vim`